



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2006
YEAR 11 MATHEMATICS EXTENSION
HSC Task #1

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Start each new question in a separate answer booklet.
- Marks may **NOT** be awarded for untidy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 76

Examiner: *D.McQuillan*

Total marks – 76

All question are NOT of equal value

Answer each QUESTION in a SEPARATE writing booklet

Question One		Marks
(a)	Which of the following are NOT polynomials.	3
	(A) $x\sqrt{x} + 2x$ (B) $\sqrt{2}x^3 + 5x - 2$ (C) $2x^3 + 5x^{-2}$	
	(D) -5 (E) $3 + 7x^3 - 2x^5$ (F) $\frac{x^2 - 4}{x^2 - x - 2}$.	
(b)	For the parabola with focus $(-1, 4)$ and vertex $(-1, 0)$ find:	3
	(i) the focal length	
	(ii) equation of the directrix	
	(iii) equation of the parabola	
(c)	Show that $x + 1$ is a factor of $x^3 - 5x^2 + 3x + 9$.	2
(d)	(i) Write down the expansion of $\sin(A + B)$.	3
	(ii) Hence find the exact value of $\sin(105^\circ)$.	
(e)	If $\tan \alpha = \frac{2}{3}$ and $\tan \beta = \frac{1}{5}$ find the value of $\tan(\alpha - \beta)$.	2

- (f) Given two polynomials, $P(x)$ of degree m , $Q(x)$ of degree n
and $\frac{m}{n} > 1$. Find the degree of: 2

(i) $P(x) \times Q(x)$

(ii) $Q(x) - P(x)$

- (g) Simplify $\sin A \sin 2A + \cos A \cos 2A$. 2

- (h) By expressing $6x^3 + 25x^2 + 5x - 1$ in the form
 $(3x + 2)Q(x) + R(x)$ find the polynomials $Q(x)$ and $R(x)$.
Given that $R(x)$ is a constant. 2

End of Question One

START A NEW WRITING BOOKLET

Question Two**Marks**

- (a) Sketch the graphs of the following equations showing all x and y -intercepts. 4
- (i) $y = (x - 1)^3$
- (ii) $y = (x + 4)^2(x - 1)(5 - x)$
- (b) The circles with centres $(-3, -2)$ and $(5, 4)$ and radii 7 and 3 respectively, intersect at only one point. By considering the division of an interval into a ratio find the point of intersection. 2
- (c) When $x^4 - 8x^3 - mx^2 - (m - 4)x - 27$ is divided by $x - 2$ the remainder is -7 . Find the value of m . 2
- (d) Find an equation of a line with an x -intercept of 3 and makes an angle of 45° with the line $y = 2x + 5$. 2
- (e) Find the Cartesian equation of the parabola with parametric equations $x = \frac{t}{2} + 5$, $y = t^2 - 1$. 2

- (f) The roots of $2x^3 + 6x + 3 = 0$ are α, β, γ . **4**
Find the value of:

(i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

(ii) $\alpha^2 + \beta^2 + \gamma^2$

- (g) Express $5x^3 - 15x^2 + 13x - 1$ in the form $A(x-1)^3 + B(x-1) + C$ **3**

End of Question Two

START A NEW WRITING BOOKLET

Question Three

Marks

- (a) Jamie and Adam want to secure a vertical pole by attaching guy wires to the top. The pole is due north of Adam and the angle of elevation to the top is $15^{\circ}18'$, this point is where the first wire will be grounded. Adam now walks 68m due west and finds the pole has a bearing 067° , this point is where a second wire is to grounded. Find the length of wire they will need for these two wires, rounded up to the nearest metre. 3
- (b) The points $P(6p, 3p^2)$ and $Q(6q, 3q^2)$ lie on the parabola $x^2 = 12y$. 6
- (i) Show that the equation of the tangent at P is $y = px - 3p^2$.
- (ii) Hence write down the equation of the tangent at Q .
- (iii) Find the point of intersection of the two tangents.
- (c) Prove the identity $\frac{\sin \phi}{1 - \cos \phi} = \operatorname{cosec} \phi + \cot \phi$. 3
- (d) In a trapezium $ABCD$, AB is parallel to DC , $AB = BC = 10\text{cm}$, $CD = 7\text{cm}$ and $DA = 8\text{cm}$. Find the size of $\angle BCD$ to the nearest minute. 2

(e) Given the expression $6 \sin x + 2\sqrt{3} \cos x$.

6

(i) Write the expression in the form $R \sin(x + \alpha)$ where $R > 0$ and α is acute.

(ii) Hence solve $6 \sin x + 2\sqrt{3} \cos x = 2\sqrt{6}$, for $0 \leq x \leq 2\pi$.

(iii) Write down the general solution for the equation in part (ii).

(iv) Find the greatest value of $6 \sin x + 2\sqrt{3} \cos x$.

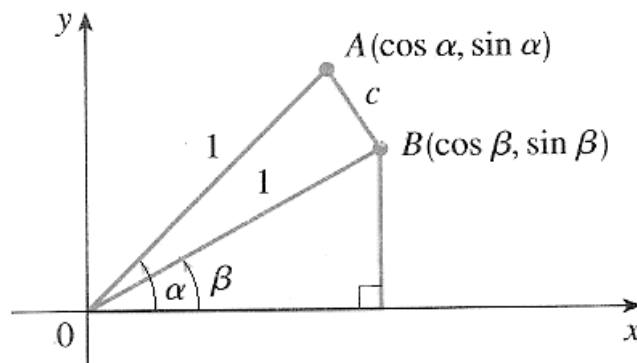
End of Question Three

Question Four

Marks

- (a) Show that $\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right) = \sec \theta + \tan \theta$. 3

- (b) Given the figure, 6



- (i) Prove the subtraction formula

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$
 by comparing two expressions for c^2 .
- (ii) Using part (i) and the fact that $\cos \theta$ is an even function and $\sin \theta$ is an odd function, deduce the addition formula for cosine.
- (iii) Use the addition formula for cosine and the identities

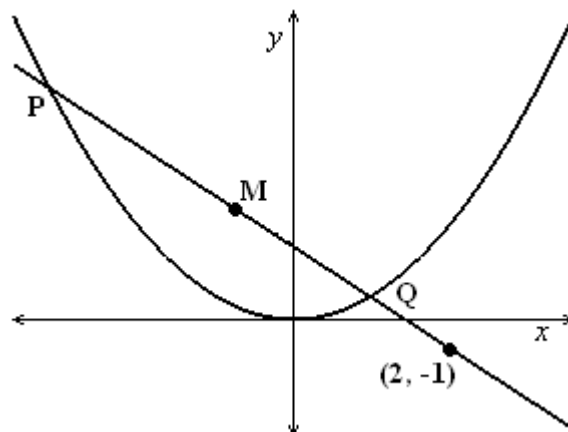
$$\cos(90^\circ - \theta) = \sin \theta \quad \sin(90^\circ - \theta) = \cos \theta$$
 to prove the subtraction formula for the sine function.

- (c) Solve $\tan x \sin x = 3 \sin x - \sec x$, for $0^\circ \leq x \leq 180^\circ$.
Round to the nearest minute.

3

- (d) The points $P\left(p, \frac{p^2}{2}\right)$ and $Q\left(q, \frac{q^2}{2}\right)$ lie on the parabola
 $x^2 = 2y$.

6



- (i) Show that the equation of the chord PQ is

$$y = \frac{(p+q)x - pq}{2}$$

- (ii) If PQ passes through $(2, -1)$ show that $pq = 2p + 2q + 2$.

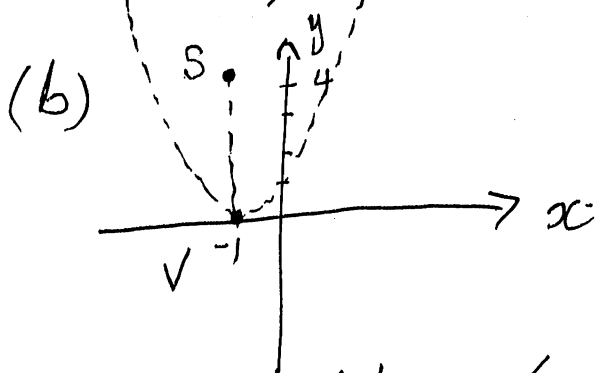
- (iii) Find the locus of M the mid-point of PQ .

End of Question Four

End of Paper

Q1/ (a) not polynomials A, C, F.

(3)



19

- (i) focal length 'a' = 4 (1)
 (ii) eqⁿ directrix $y = -4$ (1)
 (iii) eqⁿ parabola $(x-h)^2 = 4a(y-k)$
 $(h,k) = (-1, 0)$ $(x+1)^2 = 4 \times 4(y-0)$
 $(x+1)^2 = 16y$ (1)

(c) could use long division
 or let $P(x) = x^3 - 5x^2 + 3x + 9$
 let $x+1=0 \Rightarrow x=-1$

So $P(-1) = -1^3 - 5(-1)^2 + 3(-1) + 9$

$= -1 - 5 - 3 + 9 = 0$ Is a factor! (2)

(d) (i) $\sin(A+B) = \sin A \cos B + \cos A \sin B$ (1)

(ii) $\sin 105^\circ = \sin(60^\circ + 45^\circ) = \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ$

$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}}$

$= \frac{(\sqrt{3}+1)}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{4}$ (2)

$$\begin{aligned}
 (e) \quad \tan(\alpha - \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\
 &= \frac{\frac{2}{3} - \frac{1}{5}}{1 + \frac{2}{3} \times \frac{1}{5}} = \frac{\frac{7}{15}}{1 + \frac{2}{15}} = \frac{7}{17} \quad (2)
 \end{aligned}$$

(f) $P(x)$ degree m
 $Q(x)$ degree n

$$\frac{m}{n} > 1 \Rightarrow m > n$$

(i) add indices $m+n$ (1)

(ii) bigger index stays m (1)

$$(g) \quad \cos(x-y) = \cos x \cos y + \sin x \sin y$$

$$\text{So } \cos A \cos 2A + \sin A \sin 2A$$

$$= \cos(A-2A)$$

$$= \cos(-A)$$

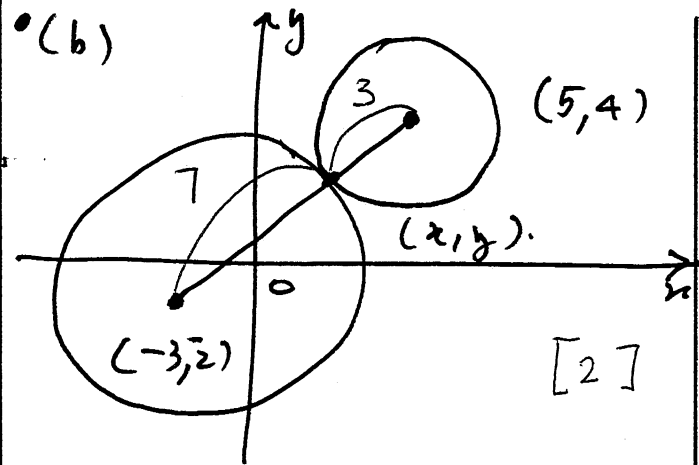
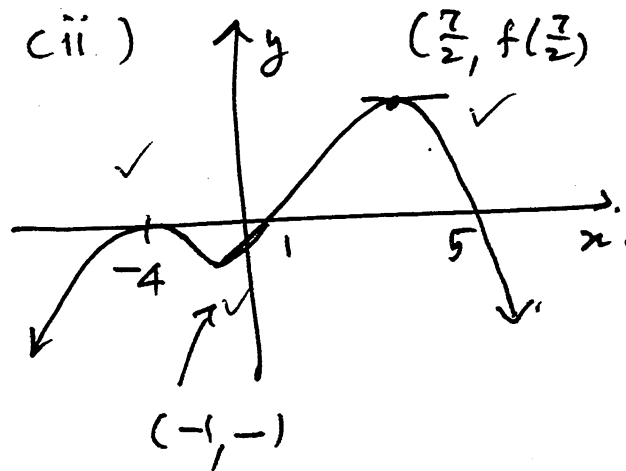
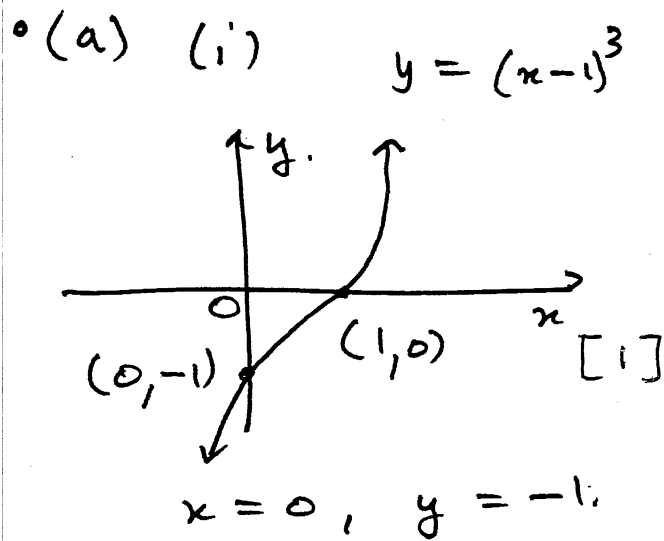
$$= \cos A \text{ even fn. } (2)$$

$$\begin{array}{r}
 (h) \quad 3x+2 \overline{) 6x^3 + 25x^2 + 5x - 1} \\
 \underline{6x^3 + 4x^2} \\
 21x^2 + 5x - 1 \\
 \underline{21x^2 + 14x} \\
 -9x - 1 \\
 \underline{-9x - 6} \\
 5
 \end{array}$$

$$(3x+2)(2x^2+7x-3)+5 \quad (2)$$

Solution Question [2]

[19]



$$x = \frac{mx_2 + nx_1}{m+n}$$

$$= \frac{7 \times 5 + (3 \times -3)}{10}$$

$$= \frac{35 - 9}{10}$$

$$2\frac{3}{5} = \frac{13}{5} = 2.6$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$= \frac{7 \times 4 + (3 \times -2)}{10}$$

$$2\frac{1}{5} = \frac{11}{5} = 2.2$$

$\therefore$ The pt is $(2.6, 2.2).$

• (c) $P(x) = x^4 - 8x^3 - mx^2 - (m-4)x - 27.$

$P(2) = -7.$ [2]

$P(2) = 16 - 64 - 4m - 2m + (-19)$

$\therefore -6m - 67 = -7$

$-6m = 60, m = -10$

• (d) $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|.$

Now $1 = \left| \frac{m - 2}{1 + 2m} \right|$

$\therefore |1 + 2m| = |m - 2|$ [2]

$\checkmark 1 + 2m = m - 2$ or $\checkmark 1 + 2m = 2 - m$

$m = -3$ $3m = 1$

$m = \frac{1}{3}$

$y = -3x + b$

$0 = -9 + b \therefore y = \frac{x}{3} + b.$

$\therefore b = 9$

• $y = -3x + 9$ | $y = \frac{x}{3} - 1$

$3x + y - 9 = 0$ | $x - 3y - 3 = 0.$

• (e) $x = \frac{k}{2} + 5$

$\therefore k = 2(x-5)$ ——— ①

$y = k^2 - 1$ [2]

$\therefore \boxed{y = 4(x-5)^2 - 1}$

or $= 4(x^2 - 10x + 25) - 1$

$\boxed{y = 4x^2 - 40x + 99}$

• (f) $2x^3 + 6x + 3 = 0$ [4]

(i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

$= \frac{3}{-3/2}$

$= -2$

2

(ii) $(\alpha^2 + \beta^2 + \gamma^2) = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 0 - 2(3)$
 $= -6$ 2

• (g) $5x^3 - 15x^2 + 13x - 1 \equiv A(x-1)^3 + B(x-1) + C$

(i) Equate coefficient of x^3 [3]

$\therefore \boxed{A = 5}$ 1

(ii) Put $x = 1$,

$5 - 15 + 12 = C \therefore \boxed{C = 2}$ 1

(iii) put $x = 0$

$-A - B + C = -1$

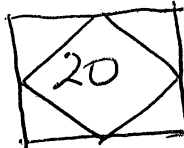
$-5 - B + 2 = -1$

$\boxed{B = -2}$ 1

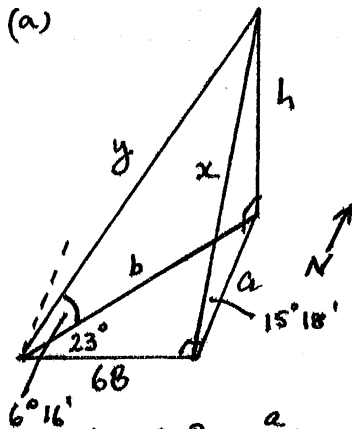
$\therefore 5x^3 - 15x^2 + 13x - 1$

$= 5(x-1)^3 - 2(x-1) + 2$

Question 3



(a)



$$\tan 23^\circ = \frac{a}{68}$$

$$a = 68 \tan 23^\circ$$

$$\cos 15^\circ 18' = \frac{a}{x}$$

$$\therefore x = \frac{\cos 15^\circ 18'}{\cos 23^\circ} = \frac{68 \tan 23^\circ}{\cos 15^\circ 18'} = 29.9249 \dots$$

$$\cos 23^\circ = \frac{68}{b}$$

$$b = \frac{68}{\cos 23^\circ}$$

$$\cos 6^\circ 16' = \frac{b}{y}$$

$$y = \frac{b}{\cos 6^\circ 16'} = \frac{68}{\cos 23^\circ \cos 6^\circ 16'} = 74.3165 \dots$$

$$\therefore x+y = 104.241 \dots = 105 \text{ m (rounded up)}$$

3

(b) (i) $y = \frac{1}{12}x^2$
 $y' = \frac{1}{6}x$

At P, eqn of tangent is

$$y - 3p^2 = p(x - 6p)$$

$$y - 3p^2 = px - 6p^2$$

$$y = px - 3p^2$$

3

(ii) $y = qx - 3q^2$

1

(iii) $px - 3p^2 = qx - 3q^2$

$$(p-q)x = 3p^2 - 3q^2$$

$$(p-q)x = 3(p-q)(p+q)$$

$$x = 3(p+q)$$

$$y = p(3(p+q)) - 3p^2 = 3pq$$

pt of n is $(3(p+q), 3pq)$

2

6

(c) $\text{RHS} = \frac{1}{\sin \phi} + \frac{\cos \phi}{\sin \phi}$
 $= \frac{1 + \cos \phi}{\sin \phi}$
 $= \frac{(1 + \cos \phi)(1 - \cos \phi)}{\sin \phi (1 - \cos \phi)}$
 $= \frac{1 - \cos^2 \phi}{\sin \phi (1 - \cos \phi)}$
 $= \frac{\sin^2 \phi}{\sin \phi (1 - \cos \phi)}$
 $= \frac{\sin \phi}{1 - \cos \phi}$
 $= \text{LHS as required.}$

3

(d) $\frac{x}{x+8} = \frac{y}{y+10} = \frac{7}{10}$

$$10x = 7x + 56$$

$$3x = 56$$

$$x = \frac{56}{3}$$

$$10y = 7y + 70$$

$$3y = 70$$

$$y = \frac{70}{3}$$

$$\cos \theta = \frac{7^2 + (\frac{70}{3})^2 - (\frac{56}{3})^2}{2 \times 7 \times \frac{70}{3}}$$

$$= \frac{3}{4}$$

$$\therefore \theta = 41^\circ 25'$$

$$\therefore \angle BCD = 138^\circ 35'$$

2

(e) (i) $6 \sin x + 2\sqrt{3} \cos x$
 $= 4\sqrt{3} \left(\frac{6}{4\sqrt{3}} \sin x + \frac{2\sqrt{3}}{4\sqrt{3}} \cos x \right)$
 $= 4\sqrt{3} \left(\sin x \cdot \frac{\sqrt{3}}{2} + \cos x \cdot \frac{1}{2} \right)$
 $= 4\sqrt{3} \left(\sin x \cos \frac{\pi}{6} + \cos x \sin \frac{\pi}{6} \right)$
 $= 4\sqrt{3} \sin \left(x + \frac{\pi}{6} \right)$

2

(ii) $4\sqrt{3} \sin \left(x + \frac{\pi}{6} \right) = 2\sqrt{6}$

$$\sin \left(x + \frac{\pi}{6} \right) = \frac{1}{\sqrt{2}}$$

$$x + \frac{\pi}{6} = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

$$x = \frac{\pi}{12} \text{ or } \frac{7\pi}{12}$$

2

(iii) General solutions $\frac{\pi}{12} + 2n\pi$ and $\frac{7\pi}{12} + 2n\pi$

1

(iv) Max value when $\sin \left(x + \frac{\pi}{6} \right) = 1$
 $\therefore \text{Max value} = 4\sqrt{3}$

1

6

Yr 11 Ext 1 Task #1 (06)

Question 4

(a) LHS = $\frac{\tan \frac{\pi}{4} + \tan \frac{\theta}{2}}{1 - \tan \frac{\pi}{4} \cdot \tan \frac{\theta}{2}}$

$$= \frac{1+t}{1-t} \quad \text{where } t = \tan \frac{\theta}{2}$$

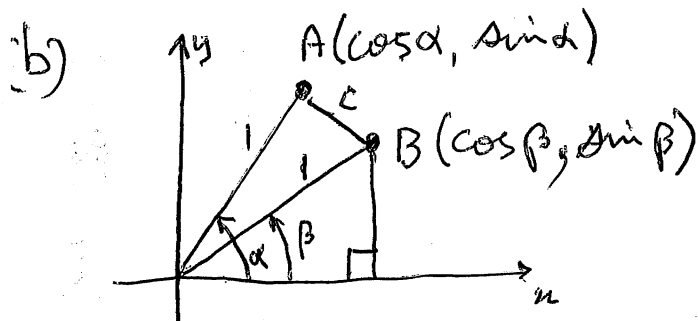
$$= \frac{(1+t)^2}{1-t^2}$$

$$= \frac{1+t^2+2t}{1-t^2}$$

$$= \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2}$$

$$= \sec \theta + \tan \theta$$

$$= \text{RHS} \quad [3]$$



(i) Distance formula

$$c^2 = (\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2$$

$$= 2 - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta)$$

By the Cosine Rule

$$c^2 = 1 + 1 - 2 \cos(\alpha - \beta)$$

$$\therefore 2 - 2 \cos(\alpha - \beta) = 2 - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta)$$

$$\therefore \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

[2]

(ii) Replace β with $-\beta$

$$\therefore \cos(\alpha - (-\beta)) = \cos \alpha \cos(-\beta) + \sin \alpha \sin(-\beta)$$

$$\therefore \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

(cosine is even, sine is odd) [2]

(iii) $\sin(\alpha - \beta) = \cos(90 - (\alpha - \beta))$

$$= \cos((90 - \alpha) + \beta)$$

$$= \cos(90 - \alpha) \cos \beta - \sin(90 - \alpha) \sin \beta$$

$$\therefore \sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

[2]

(c) $\tan x \sin x = 3 \sin x - \sec x$
($\cos x \neq 0$)

$$\frac{\sin^2 x}{\cos x} = 3 \sin x - \frac{1}{\cos x}$$

$$\therefore \sin^2 x = 3 \sin x \cos x - 1$$

$$\therefore \sin^2 x = 3 \sin x \cos x - \cos^2 x - \sin^2 x$$

$$\therefore 2 \sin^2 x - 3 \sin x \cos x + \cos^2 x = 0$$

$$(2 \sin x - \cos x)(\sin x - \cos x) = 0$$

$$\therefore 2 \sin x - \cos x = 0 \text{ or } \sin x - \cos x = 0$$

$$\tan x = \frac{1}{2} \quad \text{or} \quad \tan x = 1$$

$$x \doteq 26^\circ 34', x = 45^\circ \quad [3]$$

(d) $P(p, \frac{p^2}{2}), Q(q, \frac{q^2}{2})$ lie on

$$x^2 = 2y$$

(i) Chord PQ:

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y - \frac{p^2}{2}}{\frac{q^2}{2} - \frac{p^2}{2}} = \frac{x - p}{q - p}$$

$$\frac{2(y - \frac{p^2}{2})}{q^2 - p^2} = \frac{x - p}{q - p}$$

$$\therefore 2y - p^2 = (x - p)(q + p) \\ = (p + q)x - p^2 - pq$$

[2] $\therefore y = \frac{1}{2}((p + q)x - pq)$

(ii) Substitute (2, -1):

$$-1 = \frac{1}{2}(2p + 2q - pq)$$

$$-2 = 2p + 2q - pq$$

[2] $\therefore pq = 2p + 2q + 2$

(iii) For m:

$$x = \frac{p + q}{2} ; y = \frac{\frac{p^2}{2} + \frac{q^2}{2}}{2} \\ = \frac{p^2 + q^2}{4}$$

$$\therefore p + q = 2x \quad p^2 + q^2 = 4y$$

$$\text{Now } p^2 + q^2 = (p + q)^2 - 2pq$$

$$\therefore 4y = (2x)^2 - 2(2p + 2q + 2) \\ = 4x^2 - 2(2(2x) + 2) \\ = 4x^2 - 8x - 4$$

$$\therefore \text{Locus: } y = x^2 - 2x - 1$$

[2]